

Philosophy of Space and Time

Lecture 5: *Newton and Absolute Space*

Chris Smeenk

Department of Philosophy, UCLA · April 14, 2026

Today's Plan

- 1 Introduction
- 2 From Aristotle to Newton
- 3 Galilean Relativity
- 4 Newtonian Physics
- 5 Arguments from *De Gravitatione*
- 6 The Scholium on Space and Time
- 7 Summary and Preview

Part I: Geometry

- Zeno: puzzles about continuous motion
- Euclid → non-Euclidean geometries
- Poincaré and Reichenbach: geometry neither *a priori* nor purely empirical
- The entanglement of geometry and physics

Part II: Space, Time, Motion

- What *is* space? What structures must it have?
- Newton: physics requires various structures
- The absolute vs. relational debate
- **Today:** Galilean relativity, inertia, and Newton's way of reconciling the two

The Central Tension

Two compelling new ideas about motion, apparently in tension with one another:

Relativity of Motion

Motion described relative to some choice of bodies taken to be at rest. Different choices are equally good, lead to the same physical description.

The Law of Inertia

There is a privileged kind of motion (uniform speed, in a straight line) that requires no cause.

If motion is relative, how can the law of inertia single out one kind of motion as special?

Natural to Inertial Motion

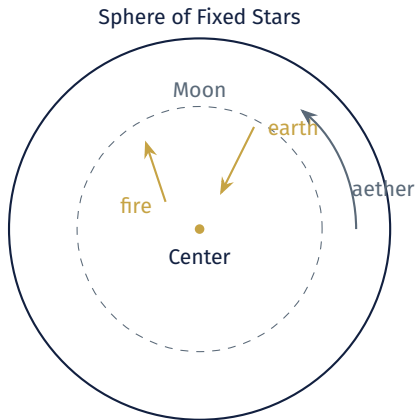
Aristotle's Universe

For Aristotle, the **nature** of an object is revealed by how it moves when left to its own devices:

- Earth falls downward (toward the center)
- Fire rises upward
- Aether moves in uniform circles

Different *kinds* of matter have different *natural motions*.

The center of the universe is a physically privileged place.



What's Wrong with Aristotle's Picture?

- Different laws for different kinds of matter—no unified physics
- The center of the universe is a privileged, absolute location
- Circular motion is “natural” for celestial matter—needs no cause
- Once you give up a finite, spherical cosmos, there is no longer a center to fall toward

?? *If there is no “center of the universe,” what takes its place as the reference point for describing motion?*

Newton's First Law

“Every body perseveres in its state either of rest or of uniform motion in a straight line, except insofar as it is compelled to change its state by impressed forces.”

— Newton, *Principia*, Law I

Maudlin: this single law “smashes the Aristotelian universe to smithereens.”

- Governs every body—stones and planets alike
- *No natural motion* tied to substance types
- Straight-line, uniform motion is the default—any deviation requires a **force**
- Circular motion (even celestial) is *not* natural; it requires a force

Galilean Relativity

Galileo's Ship

“Shut yourself up with some friend in the main cabin below decks on some large ship... Have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still.”

— Galileo, *Dialogue Concerning the Two Chief World Systems* (1632)

Galileo lists experiments below deck: dripping water, jumping, throwing a ball, watching butterflies and fish. **None of them reveal whether the ship is in motion.**

The Principle of Relativity

Galileo's thought experiment suggests a general principle:

Galilean Relativity

The outcome of any mechanical experiment is the same whether performed in a laboratory at rest or in a laboratory moving with **uniform velocity** in a straight line.

- No experiment performed “below decks” can distinguish rest from uniform motion
- The laws of mechanics take the same form in all **inertial frames**
- This is not just a practical limitation—it reflects something deep about the structure of the physical world

NB: Newton proves this as Corollary V of the Laws of Motion.

Restricted Relationalism

Notice the qualification: **uniform velocity in a straight line.**

- If the ship accelerates, turns, or rocks—you *can* tell. (Coffee sloshes, pendulums swing, you lose your balance, ...)
- Galilean relativity applies only to *inertial* observers: those not subject to accelerations
- Acceleration, unlike uniform velocity, has detectable physical consequences

?? *It is tempting to draw a stronger conclusion: that all motion is relative: that it only makes sense to speak of a body's motion relative to other bodies. But is this stronger claim warranted?*

The Temptation of Full Relationism

The idea that “all motion is relative” is seductive:

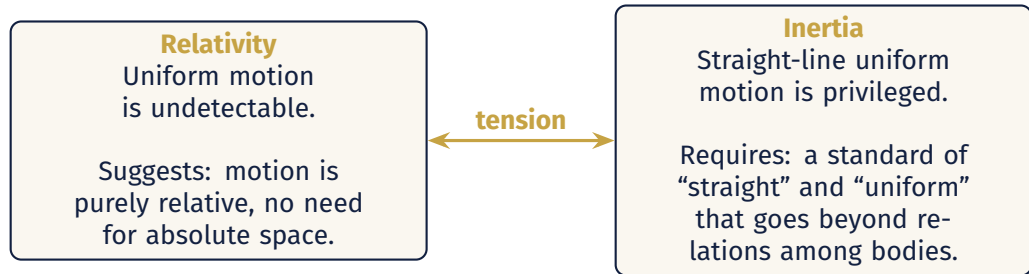
- It seems to follow from how we naturally think of motion (... motion relative to something taken at rest)
- It seems ontologically parsimonious (no need for invisible “absolute space”)
- Descartes embraced it: motion is “transfer from the vicinity of contiguous bodies”; we will see Leibniz’s version later

The Temptation of Full Relationism, 2

But there is a problem. Once we accept **inertia** we need answers to questions that a relationalist struggles to provide:

- Straight with respect to *what*?
- Uniform give *what* way of measuring time?
- How do we distinguish a body that is genuinely accelerating from one that merely *appears* to accelerate because nearby bodies have moved?

The Conflict: Relativity vs. Inertia



This tension is the driving problem for the seventeenth century debates .
Newton's response: embrace *absolute space* to support inertia, while also accepting that absolute velocity is undetectable (for more, see also Belot 2000, "Geometry and Motion").

Newtonian Physics

Inertia Demands Structure

The First Law codifies **inertia**: a body free of forces continues in uniform straight-line motion.

But stating this law precisely requires answering the following questions:

- What counts as a “straight line” in space?
- What counts as “uniform” motion over time?
- What does it mean to “remain at rest”?

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Newton's answer to all three: features of **absolute space** and **absolute time**.

Two sources for Newton's views:

- *De Gravitatione* (unpublished, ca. 1684): extended philosophical analysis
- The *Scholium* to the Definitions in the *Principia* (1687): compact and precise

What Does the First Law Require?

The Law

A body with no net force maintains the same state of motion: uniform motion in a straight line. (Change in state of motion requires a force.)

Unpack what is required (following Maudlin, Ch. 1):

1. What does “straight line” mean? → space must have **affine structure**
2. What does “uniform” mean? → a **metric** on space and a **metric** on time
3. What does “remains at rest” mean? → points of space must **persist through time**

Each requirement corresponds to a layer of geometric structure that Newton needs.

Layers of Structure

Topology: which points are “nearby”—continuous paths

for “continuous motion”



Affine structure: which paths are “straight”

for “straight line”



Metrics: distances between points, times

for “constant velocity”



Persistence of space through time: same point at different times

for “at rest”

Affine Structure: Straight vs. Curved

The **affine structure** of space determines which paths count as straight lines. (Paths whose tangent vector remains parallel.)

- In Euclidean space E^3 , straight lines are the familiar ones
- On a sphere, “straight lines” (geodesics) are great circles
- (Generalizes to other, more complicated surfaces, and to spacetime.)

Newton's First Law says: without forces, bodies travel along straight lines through space (*and time*).

Euclidean: straight



Not straight: force required



Great circle on sphere

The Persistence of Space

?? What does it mean for a body to be “at rest”?

For “at rest” to make sense, need some way to identify “same points of space” at different times.

- A body is at **absolute rest** when it occupies the same points of absolute space over time
- This requires that **space** has an identity through time—its parts *persist*
- Otherwise, “staying in the same place” is meaningless

This is the most controversial of Newton’s assumptions, denied by relationalists.

Absolute Time

The First Law also requires **absolute time**:

- “Uniform” motion = equal distances in equal times
- This presupposes a *metric on time*: a fact about how much time elapses between any two instants
- Clocks *measure* time, but time itself is independent of any particular clock

“Absolute, true, and mathematical time, of itself, and from its own nature, flows equably without relation to anything external.”

— Newton, *Scholium*

Break

De Gravitatione: Against Descartes

De Gravitatione: Context

Newton's *De Gravitatione* (unpublished, ca. 1684) is an extended philosophical treatise, (probably) written as Newton was developing the *Principia*.

- Its primary target is Descartes' *Principles of Philosophy* (1644)
- Descartes defined motion as “transfer from the vicinity of contiguous bodies”—a purely relational definition
- Newton argues that this definition is *incoherent*: it cannot support the very laws of motion Descartes himself proposed

This is not a minor complaint; it directly targets the tension between relativity and inertia.

Newton's Critique: No Definite Motion

Newton's objections to Descartes' definition:

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“It follows indubitably that Cartesian motion is not motion, for it has no velocity, no determination, and there is no space or distance traversed by it.”

— Newton, *De Gravitatione*

Newton's Critique: Descartes' Self-Contradiction

Newton presses the point further:

- Descartes' *own* laws of motion require a notion of “tendency to rectilinear motion” (Descartes' Law II) and change in speed due to collision (Law III)
 - But on Descartes' own *definition* of motion, these concepts are meaningless—there is no fact about whether a body is moving in a straight line or a curve
 - Descartes seems to know this: he quietly shifts between his official relational definition and an unofficial appeal to absolute motion when stating his laws
- ?? *Newton's objection is not just that Descartes gets the physics wrong, but that Descartes' definition of motion is incompatible with his own physics.*

Newton's Positive Conclusion

"...[I]t is necessary that the definition of places, and hence of local motion, be referred to some motionless being such as extension alone or space in so far as it is seen to be truly distinct from bodies."

— Newton, *De Gravitatione*

Motion must be defined with respect to an independently existing, **immovable** space.

- Space is not a substance (like matter) nor an accident (like color)
- It has a unique status: it "approaches more nearly the nature of substance"
- Space is **absolute**: it does not depend on bodies for its existence
- It is **immutable**: its geometrical structure does not change
- It is **infinite** in all directions, with Euclidean structure

Newton's Conception of Body

In *De Gravitatione*, Newton also offers a remarkable account of *matter*:

- Contra Descartes: matter $\neq$ extension. Space can exist without bodies.
- Newton's thought experiment: God could endow regions of space with certain properties—mobility, impenetrability, the ability to excite sensation
- Body = a region of space that God endows with force and then moves according to laws

?? *On this account, bodies are in a sense structured regions of space. Note how fundamental Newton's account of space is to the rest of his physics.*

**The Scholium:
By Properties, Causes,
and Effects**

The Scholium: Overall Structure

The Scholium to the Definitions in the *Principia* is Newton's most compact statement of his views. Structure:

§§I–VI: Introduction of the distinctions between absolute and relative space, time, place, and motion

§VIII–XIII: Defense of these distinctions “by their **properties**, **causes**, and **effects**”

§XIV: The epistemological problem: how to determine true motions from observed (relative) motions

We will follow this structure today and continue with the “effects” argument (the bucket) on Thursday.

Absolute vs. Relative: The Distinctions (§§I–VI)

Newton carefully separates:

Absolute

- Absolute space: immovable, the same always and everywhere
- Absolute time: flows equably, without relation to anything external
- Absolute motion: change of position in absolute space; *unique*

Relative

- Relative space: a moving frame (the hold of a ship, the air around us)
- Relative time: sensible measures (clocks, planetary periods)
- Relative motion: change of position relative to bodies; *multiple*

The relative quantities are what we *observe*. The absolute quantities are what the *laws* refer to.

Newton's Ship Example

“If the earth is really at rest, the body that relatively rests in the ship will really and absolutely move... But if the earth also moves, the true and absolute motion of the body will arise...”

— Newton, *Scholium*

- A sailor below deck is at *relative* rest with respect to the ship. But the ship moves on the sea, the sea moves with the earth, the earth moves in absolute space, ...
- True motion = motion through *absolute* space
- A body has many relative motions **but only one absolute motion**

By Properties (§§VIII–X)

Newton identifies three **properties** of absolute motion that cannot be captured in terms of relative positions:

- §VIII: Bodies truly at rest are at rest *with respect to one another*. But “true rest” cannot be determined from position relative to local bodies—those bodies may themselves be in motion.
- §IX: If a body is at rest relative to a system that is itself truly moving, then that body is truly moving too. The motion of interior parts participates in the motion of the whole.
- §X: A place that is itself in motion makes “relative motion” depend on the motion of the place—leading to an infinite regress unless there is an immovable reference: absolute space.

Causes of Absolute vs. Relative Motion

An impressed force is **necessary and sufficient** for changing a body's true state of motion. But it is **neither necessary nor sufficient** for generating a relative motion.

- You can change a body's relative motion without any force on *that* body—just move the reference body
 - Conversely, you can apply a force to a body and change its true motion while its relative motion to certain other bodies stays the same (if those bodies accelerate equally)
- ??** *The causal argument: forces are real; they produce real changes. The changes they produce track absolute motion, not relative motion.*

By Effects (§XII): Preview

Newton's most famous argument:

The Bucket Experiment

The existence of centrifugal effects (water curving away from the axis of rotation) is neither a necessary nor sufficient condition for the existence of *relative* rotation. It *is* a necessary and sufficient condition for *absolute* rotation.

We will work through the bucket experiment in detail on **Thursday**, along with the **globes** thought experiment and Newton's epistemological strategy (§XIV).

Maudlin (Ch. 2) develops this argument and its implications at length.

The Epistemological Problem (§XIV)

Newton is acutely aware of a difficulty:

- The laws of motion refer to absolute quantities (position, velocity, acceleration in absolute space)
- But we can only *observe* relative quantities (positions and motions of bodies relative to one another)
- Absolute space is imperceptible

“It is indeed a matter of great difficulty to discover, and effectually to distinguish, the true motions of particular bodies from the apparent.”

— Newton, Scholium

Newton's strategy: infer absolute motions from their observable **effects**—from the forces that accompany them.

Newton's Empirical Strategy

How does Newton propose to get at absolute motions?

1. Observe the *relative* motions of bodies
2. Infer the *forces* that produce those motions (using the laws of motion)
3. Use the forces to determine the *true* (absolute) motions

This is the program of Book III of the *Principia*: from the observed motions of the planets, infer gravitational forces, then determine the true motions of the solar system (up to the motion of the center of gravity).

?? *Is this circular? Newton thinks not: inconsistencies between inferred forces and observed motions would reveal errors.*

Summary

1. The **relativity** of uniform motion (Galileo's ship) suggests that motion might be purely relative—but the **law of inertia** demands a standard of “straight” and “uniform” that goes beyond relations among bodies
2. In *De Gravitatione*, Newton shows that Descartes' relational definition of motion makes the laws of motion unstable
3. Newton recognized that physics needs **absolute space** and **absolute time**: independently existing structures, with respect to which motion is defined
4. In the Scholium, Newton defends the absolute/relative distinction by their **properties, causes, and effects**
5. The epistemological problem: absolute space is imperceptible, so true motions must be inferred from forces

For Thursday

- **Read:** Maudlin, Ch. 2; DiSalle (C 5); Geroch, Chs. 1–2.
- **Think about:**
 - › What does the bucket experiment show, exactly? What is it evidence *for*?
 - › How does Newton's “effects” argument relate to the tension between relativity and inertia?

Thursday: **The Bucket, the Globes, and the Leibniz-Clarke Debate**

- Newton's experimental arguments for absolute motion
- The Leibniz-Clarke debate and the shift arguments
- What structure does physics *really* need?

Overflow: Galileo's Circular Inertia

Before Newton, Galileo had argued that the “inertial” motion of terrestrial objects would be uniform *circular* motion around the earth.

- Galileo's inclined plane experiments: a ball rolls up to the same height it started at
- Inertial motion: ball rolling such that it stays at the same height (neglecting friction)
- But “forever on a flat plane at equal distance to the center” = circular motion around the center

Newton's breakthrough: *straight-line* motion, not circular. Circular motion requires a force (gravity).

Overflow: Corollary V—Newton's Relativity Principle

"The motions of bodies included in a given space are the same among themselves, whether that space is at rest, or moves uniformly forwards in a right line without any circular motion."

— Newton, *Principia*, Corollary V

Newton himself recognized that *uniform* motion through absolute space is undetectable.

- Absolute *velocity* is unobservable
- Only absolute *acceleration* (including rotation) has observable effects
- This will become important when we ask: does Newton's physics really need *all* the structure of absolute space?

Questions?

chris.smeenk@gmail.com